

THE MID INSTRUMENT OF EUROPEAN XFEL: 2025 UPGRADES AND EXPERIMENTAL SETUPS

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Abstract

In this paper insights are given on some examples of 2025 Upgrades and Experimental Setups at the MID instrument of European XFEL.

The latest design and implementation status of **MID Multi-Purpose Chamber (MPC-2)** project are reported. This represents a set of upgrades and ongoing development of MID instrumentation, as well as the evolution of the current multi-purpose chamber, which has been successfully used at MID since the first experiments in 2019. The aim is to enable new types of scientific experiments and to expand the current capabilities. The MPC-2 project includes an upgrade of the MPC-2 VESSEL and the MPC-2 INTERIOR Upgrade (MPC-2_IU). The latter consists of the following assemblies: Breadboard Assembly, Laser In-coupling 2 (LIC-2), 2-theta Circle.

Another aim of the MPC-2 project is to make operation of experiments easier with better access to sample environments (Pulsed Magnet (PUMA), Cryostat, Fast Solid Sample Scanner (FSSS), Diffractometer, Liquid Injector), and possibilities to install new ancillary equipment included in the MPC-2 INTERIOR Upgrade.

Examples of the most recent EXPERIMENTAL SETUPS are also reported, continuing in the direction of simultaneous multi-detector-use.

MPC-2 VESSEL UPGRADE

A new MID Multi-Purpose Chamber 2 Vessel (MPC-2_V) is required to enable parallel in-coupling of both the optical pump (PP) laser and the X-ray beam. This configuration allows the use of the PP laser in combination with the nano-focusing optics (NaFo). The redesigned vessel supports the use of different wavelengths provided by the PP laser, enabling quick replacement of several laser windows and optics within the new LIC-2 setup during experiments, via large access doors. In Fig.1-3 the most recent concept design status, budgetary quote release candidate, is shown.

Design

During previous MPC operation, we have identified a significant benefit in improving accessibility to the sample interaction point. The design indicated in these paper and poster outlines which existing vessel features - such as feedthrough flanges - are retained, and which are obsolete. Areas where flanges are not required are instead used for

significantly increased access ports or doors. The current lid will remain available as a fallback solution.

The MPC-2_V must fulfil several key boundary requirements, including full compatibility with:

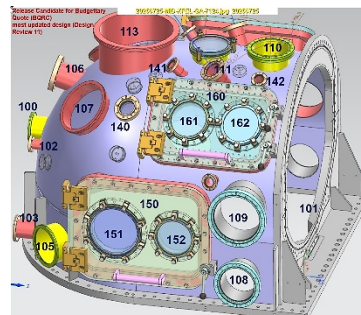


Figure 1: last version of MPC-2 Vessel Upgrade with flange numbering (North side).

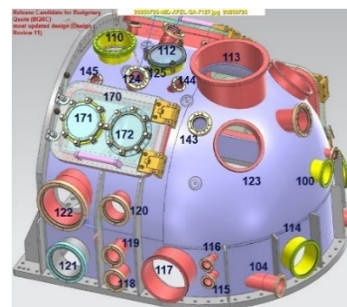


Figure 2: last version of MPC-2 Vessel Upgrade with flange numbering (North side).

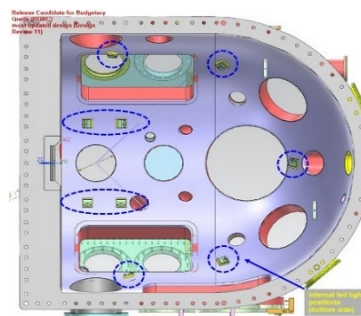


Figure 3: last version of MPC-2 Vessel Upgrade with internal led lights positions (Bottom side).

- The existing lower chamber section onto which it will be mounted, as well as with fixed positions and dimensions for X-ray and PP laser in-coupling.
- The high-vacuum generated with the existing interfaces: chamber base, WAXS window, DN500 gate valve.

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- Guaranteeing easy access to the sample interaction point and to several internal setups via larger side ports.
- The LIC-2 setup - and it should be craneable for installation without requiring disassembly of LIC-2.
- The assemblies (some in progress in parallel) of the MPC-2 INTERIOR Upgrade (MPC-2_IU).
- The current environment of the X-ray Scattering and Imaging System (XSYS) [1,2,7,8,9] and infrastructure;
- The ancillary sample environment setups, indicated in the next section.
- Other additional future upgrades (e.g. the Interferometer Project).

Although the new internal Breadboard Assembly allows for installation of cameras, certain procedures - such as in-vacuum sample exchange or precise alignment of the centre of rotation - require direct visual monitoring. Therefore, the new lid includes flanges dedicated to permanent mounting of cameras for monitoring and alignment and supports for led lights have been included in the design (see Fig. 3).

MPC-2 INTERIOR UPGRADE: ANCILLARY EQUIPMENT AND TYPICAL SAMPLE ENVIRONMENTS SETUPS

The MPC-2 project also comprises implementation of the MPC-2 INTERIOR Upgrade (MPC-2_IU), featuring the following ancillary equipments represented here (see Fig. 4 circled in orange for details):

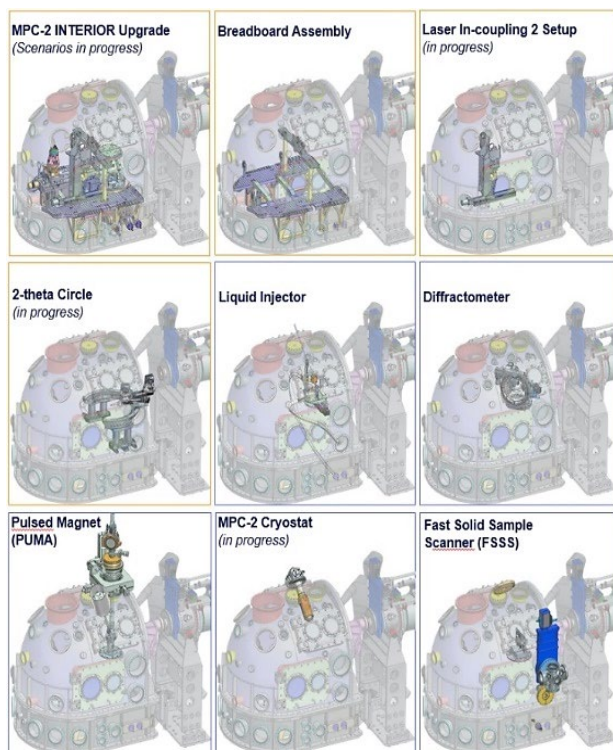


Figure 4: MPC-2 INTERIOR Upgrade Ancillary equipments (circled in orange) and typical Sample Environments (in blue).

- An upgraded Breadboard Assembly (BA): a suitable support structure designed to ensure a secure mounting of the future LIC-2, as well as mounting and aligning lenses, mirrors and cameras. It will be fixed to the existing threads at the bottom part of the current MPC.
- A new Laser In-Coupling (LIC-2): a device positioned downstream of the NaFo setup and movable along the beam direction, designed to align the X-rays and optical laser beam to hit the sample colinearly.
- A new 2-theta Circle, a device with baseplate motorized on a circular rail, to be installed on BA and used to precisely position and move additional detectors around the interaction point, e.g., ePix [4] or Jungfrau [5].
- The poster also shows the status of typical Sample Environments Setups that will be used with MPC-2: Pulsed Magnet (PUMA), Cryostat, Fast Solid Sample Scanner (FSSS), Diffractometer, Liquid Injector. (see Fig. 4, circled in blue).

EXPERIMENTAL SETUP FOR HIGH RESOLUTION DIFFRACTION STUDIES ON ELECTRON DYNAMICS

In this experiment the Bragg reflections of the charge density wave (CDW) were measured, excited by an external current source. The experiment aims to understand the dynamics of CDWs in a quasi-2D material.

Design and Implementation

This experiment was performed in vacuum and the AGIPD detector [3] was positioned ca. 8m away from the sample (high resolution). It required design and implementation of a setup (see Fig. 5) comprising several dedicated sample holders of Cu, retainers and adapter plates in order to:

- Interface with the CDW sample mounted on a sapphire substrate;
- Interface with the current manipulator setup and with the Peltier element used to control the temperature of the sample;
- Guarantee accessibility to the sample interaction point also during the experiment (i.e. breaking the vacuum);
- Guarantee electrical contacts to drive the dynamics in the sample via an applied current and a precise sample alignment.

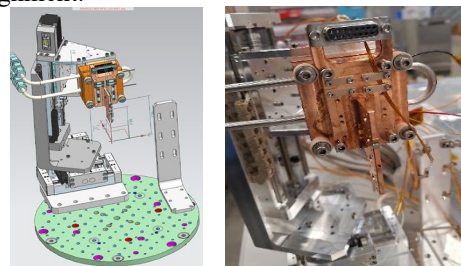


Figure 5: Design and implementation of Experimental Setup (detail)

EXPERIMENTAL SETUP FOR MEASURING BULK DISLOCATION MOBILITY WITH DARK-FIELD X-RAY MICROSCOPY

This experiment images the dynamics of dislocation/defects in a crystal after an optical laser excitation, helping to understand how dislocation influence materials properties and fatigue. The dark field microscopy magnifies the diffraction signal with additional CRL lenses and allows for spatial resolution in combination with the 1D focusing of the X-rays on the sample.

Design and Implementation

The design required the implementation of a setup (see Fig. 6) with 3 hexapodes (on which the different sets of lenses were mounted), installed on motorized linear stages, in turn mounted on the 2-theta circle, plus a diffractometer, and making use of several dedicated adapter plates.

In fact, the setup satisfied the request of a precise sample alignment of the single crystalline samples and x-ray optics to image a dislocation inside the sample.

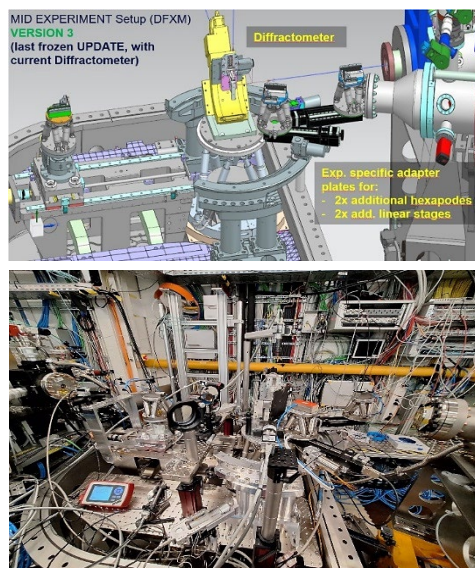


Figure 6: Design and implementation of Experimental Setup of Experimental Setup (detail)

Performed in air, the setup with the 2-theta circle was implemented in order to front the challenge of guaranteeing the precise alignment of the dark-field lenses, especially for different reflections (diffraction angles).

In addition, the setup foresaw the installation of a set of lenses to magnify in the bright field (the forward direction) for normal transmission imaging of the sample, contrast being given by the density variation in the sample.

EXPERIMENTAL SETUP FOR MEASURING TOTAL SCATTERING/PAIR DISTRIBUTION FUNCTION

Measuring the total scattering/pair distribution function (PDF) gives a high-resolution picture of for example molecules. The method requires a coverage of a large

diffraction angle range via a large detector very close to the sample. In combination with the optics for the laser execution, the liquid jet setup gives a very dense sample environment.

Design and Implementation

The design of this experimental setup required a positioning of the VAREX detector [6] close to the sample, realized via a dedicated R+K support structure, and a fixture adjustable at angles of 15 deg. and 18 deg. (see Fig. 7). The experiment was performed at high photon energies in air.

It also required the implementation of a dedicated liquid-jet catcher assembly (with 3D-printed disposable units) and of a recirculation system. Due to the hazardous nature of the sample, special caution (spray shield, exhaust connected to the catcher) had therefore to be taken.

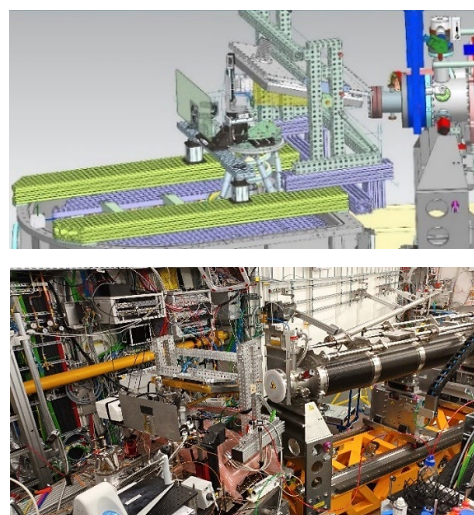


Figure 7: Design and implementation of Experimental Setup of Experimental Setup (detail)

Additionally, the total scattering signal was also recorded in combination with optical pumping as a function of pump delay.

The VAREX detector [6] was positioned in a tilted geometry above the sample to maximize the angular range of the scattering signal.

CONCLUSION

Several upgrades under the title MPC-2 project are currently underway at the MID instrument of European XFEL. The MPC-2 VESSEL design has now reached the advanced status of release candidate for budgetary quote. In parallel, several ancillary equipments of the MPC-2 INTERIOR Upgrade and Sample Environments that will be used with MPC-2 have been completed, or are in very advanced status.

Together, these devices will allow for improved operation of MID and simultaneous use of nanofocused X-ray beams and optical laser excitation of the sample.

The flexibility of the MID instrument is further illustrated by examples of experimental setups in which the MPC-2 project aims at standardizing some of these configurations.

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